4 It would take 120 minutes to fill a swimming pool using water from 5 taps.

(a) How many minutes will it take to fill the pool if only 3 of the taps are used?

..... minutes

(2)

(b) State one assumption you made in working out your answer to part (a).

.....

.....

(1)

(Total for Question 4 is 3 marks)

5 A plane travels at a speed of 213 miles per hour.

(a) Work out an estimate for the number of seconds the plane takes to travel 1 mile.

213 mph ≈ 200 mph; 1 hour = 3600 seconds

1 mark
Convert one hour to seconds

	Miles	Seconds	
	200	3600	
÷200	1	18	÷200

1 mark
Scale the time for 200 miles to 1 mile

1 mark
18 seconds is a suitable estimate

18 seconds

(3)

(b) Is your answer to part (a) an underestimate or an overestimate?
Give a reason for your answer.

Overestimate: I rounded the speed down.

A lower speed gives a longer time for the same mile.

(1)

1 mark
Reason matches the estimate

(Total for Question 5 is 4 marks)

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